

AERODYNAMICS

III Semester								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6AE04	PCC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	40	60	100
UNIT-I	FUNDAMENTALS OF FLUID MECHANICS							
Flow properties, Hydro static law, Piezometers, Simple and differential manometers, Velocity Potential and Stream function -Stream line, path line, streak line, stream tube, classification of flows, steady, unsteady, uniform, non-uniform, laminar, turbulent, rotational, irrotational flows, Continuity equation, Momentum equation -Euler's and Bernoulli's equation, Navier Stokes equation (explanation only),								
UNIT-II	ELEMENTARY FLOWS							
Introduction to Elementary Flows-Uniform flow, source flow, sink flow, doublet flow and vortex flow, non-lifting and lifting flow over a circular cylinder. Kutta- Joukowski theorem and Magnus Effect, D'Alembert's paradox								
UNIT-III	AIRFOIL AND FINITE WING THEORY							
THIN AIRFOIL THEORY: Vortex Filament, The vortex sheet, Kutta condition and Kelvin 's circulation theorem. Classical thin airfoil theory: symmetric and cambered airfoil. aerodynamic centre, centre of pressure, estimation of coefficient of lift and pitching moment.								
Finte Wing Theory: Downwash, induced drag, Biot-Savart's law and Helmholtz's theorem. Prandtl's classical lifting line theory and fundamental equations. Elliptic and general lift distribution over finite unswept wings, effect of aspect ratio, taper and thickness to chord ratio, Subsonic flow past swept and delta wings.								
Assignment/Project submission: Program to calculate lift over wing Prandtl's classical lifting line theory								
UNIT-IV	APPLIED AERODYNAMICS-I							
Flow regimes -Flow over Wing and airfoil, Lift augmentation and Drag Reduction methods - Flaps, slats, slots, winglets, Leading edge root extensions, Large Eddy Breakup device, Co-flow jet, Cuffs and vortex generators Circulation control, strakes. Critical Mach number, drag divergence Mach number, supercritical airfoils, area rule. Drag augmentation methods – spoilers, Air brakes.								
UNIT-V	APPLIED AERODYNAMICS-II							
Propellers: Airscrew geometry, Froude Momentum Theory, Thrust Co-eff, Torque Co-eff, Power Co-eff, Efficiency, Activity factor, Blade element theory.								
Boundary layer Theory:								
Boundary layers, Laminar flow and turbulent flow, Boundary layer thickness, displacement and momentum thickness, boundary layer separation.								
Assignment/Project submission: Prepare models for wings different flap positions, and leading-edge devices								
Text Books:								
1. John. M. Cimbala Yunus A. Cengel (2019), Fluid Mechanics: Fundamentals and Applications, 4 th edition, McGraw-Hill, ISBN: 9353166217								
2. Anderson J.D. (2011), Fundamental of Aerodynamics, 5 th edition, McGraw-Hill International Edition, New York.								
Reference Books:								

1. L. J. Clancy, Aerodynamics, 1/e, Shroff Publications, 2006
2. J. J. Bertin and R. Cummings, Aerodynamics for Engineers, 6/e, Pearson, 2013.

COURSE OUTCOMES: STUDENTS SHOULD BE ABLE TO:

1. Formulate and apply the continuity equation and momentum equation to solve problems related to fluid flow.
2. Apply the Kutta-Joukowski theorem to calculate lift and understand its implications for airfoil and wing design.
3. Gain proficiency in Prandtl's classical lifting line theory, including the fundamental equations, to analyze and optimize lift distribution over finite unswept wings
4. Apply principles of circulation control and strakes to manipulate airflow over wings and airfoils for enhanced performance and control.
5. Evaluate the efficiency of airscrews and assess their performance characteristics, taking into consideration factors like blade design and operating conditions.